

Important Question for DBMS

1. Explain advantages of DBMS.
2. Function and responsibility of DBA.
3. Component of Data dictionary.
4. Draw and Explain E-R model Symbol.
5. Explain NOT NULL and CHECK constraint.
6. What is Data Model? Explain all Data Model.
7. Explain Three-tier Architecture.
8. Explain Group by having and order by with Example.
9. Explain DDL and DML commands with Example.
10. Explain Data Types of DBMS.
11. Explain Generalization and Specialization with example.
12. Give the Difference between primary key and unique key.
13. Explain foreign key with example.
14. Explain different types of join with example.
15. What is sub query? Explain with Example.
16. List set operators and Explain all with Example.
17. Explain types of keys with example.
18. Explain group function with example.
19. Explain logical operators with example.
20. List types of DBMS and Explain distributed and Centralized DBMS.
21. Advantages of Three-tier architecture.
22. Explain schema, sub-schema and instance.
23. Explain Data Independence
24. What is Data Dictionary? Explain its types.
25. Define Following
 - a) Data
 - b) Information
 - c) System Catalog
 - d) Data Warehouse
 - e) Entity
 - f) Attribute
 - g) Metadata
26. Explain relational algebra operation
 - a) Selection
 - b) Projection
 - c) Intersection
 - d) Cartesian product
 - e) Rename

1. Explain advantages of DBMS.

Ans.

1. **Minimal Data Redundancy.** Due to centralized database, it is possible to avoid unnecessary duplication of information. Ex. All information about bank customer can be kept centralized.
2. **Improved Data Inconsistency.** Data inconsistency occurs due to data redundancy. Due to centralized database it is possible to avoid unnecessary duplication of information.
3. **Efficient Data Access.** DBMS provides a variety of techniques to retrieve data.
4. **Improved Data Sharing.** In DBMS Data is maintained centrally, all authorized users and application programs can share this data easily.
5. **Improved Data Integrity.** Data integrity means that data contained in the database is both correct and consistent. DBMS software provides different way to implement such types of rules.
6. **Guaranteed Atomicity.** Any operation on database must be atomic. It is responsibility of the DBMS software ensure such kind of atomicity.
7. **Improved Concurrent Access.** Multiple users are allowed to access data simultaneously (concurrently). As Database is maintained centrally, data can be shared easily among multiple users.
8. **Improved Security.** DBMS software provides way to control the access to data for different user according to their requirements.

2. Function and services of DBA.

Ans.

1. **Defining conceptual schema and database creation (Schema Definition).** The DBA Defines the overall logical structure of database. According to this schema database will be developed to stored required data for system.
2. **Storage structure and access method definition.** DBA decides how the data is to be represented in the stored database process is called physical database design.
3. **Assisting Application Programmers.** The DBA provides assistance to application programmers to develop application programs.
4. **Granting authorization to users (Approving Data Access).** The DBA determines which user access to which part of the database. This is required to prevent unauthorized access of a database.
5. **Physical organization modification.** The DBA modifies the physical organization of the database to improve performance.
6. **Monitoring performance.** The DBA monitors performance of the system and making changes in physical or logical schema if required.